

Bholenath Waghmode

Embedded Software Engineer

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2018-10
To
Current

Professional Experience

Embedded Software Engineer

Amphenol Interconnect India Pvt. Ltd., Hyderabad, India

Projects and Responsibilities

Wireless Charger (Autosar Solution)

- S32 DS IDE, multilink Debugger configuration and workspace setup.
- Board brings up.
- LIN driver configuration using EBTresos and Test, LINif software module integration.

Automotive Charger with Gesture Control:

- Elmos gestures sensor library integration with WCT2013.
- Integrated SPI driver with NXP WCT2013.
- Developed device drivers for GPIO, Timer.
- Written firmware for the Software Modules System Tick and Delay.
- Source code analysis and design document preparation using Enterprise architect tool.

Gesture Control:

- Maxim gesture sensor library Integration with ST32k144.
- CAN driver integration with S32k144.
- Designed and developed application for ST32k144 MCU to send gesture result over CAN network.
- Developed desktop application using QTC++ to show visual presentation of received gesture result over CAN network.
- Developed license free tool for CAN bus analysis (Desktop application). Its's cost cutting.
- Tools used; S32 Desing studio, QTC++, ValueCan4.

Automotive Type A+C USB /PD Charger:

- Written firmware for device configuration (Clock, GPIO, PWM, Timer) and application for EFM8F10A2A
- Released the SW for validation and Production.
- SW requirement design and preparation as per customer spec
- Developed desktop application for microchip HUB Programming.

Automotive Wireless Charger:

- Jenkins builds Integration with IAR tool.
- Designing and development of EOL Test Panel Using Vehicle Spy and Vector CANoe.
- Boot loader User Interface development Using CAPL Script and QT C++.
- LIN Conformance Test Configuration setting, Running and analysis using Vector CANoe.
- Field Issue Analysis and Root Cause finding.
- LIN driver and LIN node test case writing as per OSI layers and driver testing using Microchip LIN analyser.
- Functional software test cases writing.
- Module unit testing using Hitech, CANTATA and testidea.
- Developed Python application to automate Test cases.
- MISRA C static analysis using Scitool understand.
- Debugging using lsystem winidea and Hitech compiler.
- Module design review and approval.
- Implemented test process as per ISTQB standard.

High Lights

Core Profile

- Embedded SW Engineer

Plus points

- Hands on Firmware Development
- Hands on GUI development.
- Hands on Software Debugging and Testing.
- Hands on hardware Test and troubleshooting.

Appreciation:

- On the spot best performer award by Schinder's global R&D dept. head.

Skills

Programming Languages

- Embedded C/C++, C#, Visual Basic, Python, CAPL, Vehicle Spy Script and Assembly.

Communication Protocols

- RS232, RS485, UART, LIN,
- SPI & I2C, CAN and Ethernet.

Microcontrollers

- 8bit/16bit/32bit
- 8051/ARM
- AT89c51, AT89c2051/4051, DS222L, SST89E516, Infineon TC212, NXP WCT1013A, WCT2013, STM8S208RB, STM32F104, EFM8Busybee, S32k144, MWCT2016S
- BeagleBone Black

Tools

IDE

- Hitech, QT++, CodeWarrior (DSC), IAR, MPLAB, Visual Studio, Keil uv5, Proteus. DS32 Studio, Simplicity Studio, IAR, STVD, STM cube, EBTresos

Debugger

- Miniwigller, Multilink, IC557 (I system), Silicon labs debug adapter, ST-Link, Jink

Testing

- MPLAB CC, I system winidea, Test idea, Vector, CANoe, Vehicle Spy, uchip LIN Analyzer, Cantata, PClint, SciTool, Hercules, Tera term, Jenkins.

	Media Hub: - System testing and test case design and development in Enterprise Architect. - Unit testing using CANTATA tool. - USB Hub Programming Tool design and development using QTC++.	Other - HPQC, JIRA, IBM Rational Clear Quest, Windchill, Enterprise Architect, PRISM, Confluence, Doors
2016-12 To 2018-09	Associate Engineer Schindler India Pvt. Ltd. Pune, India Projects and Responsibilities Elevator: - Simulator designing and troubleshooting as per schematic. - Windows SW development to simulate elevator status with respect to traced log. - RS232 to TTL converter Designed, used tera term for verification. - Defect logging using JIRA, Clear Quest, and approvals in CCB meeting. - Collaboratively working with global team. - SW Validation over a simulator and test management using HpQC. - Test case design, peer review and requirement analysis. - Failure mode analysis and finding critical issues. - SW qualification report preparation.	Training/Certification - Embedded C and RTOS by Kernel Master Hyderabad. - Linux device driver Developer Level 1 by Kernel Master Hyderabad. - Linux device driver Developer Level 2 by Kernel Master Hyderabad. - Vector CANoe CAN Fundamental - Vector CANoe LIN Fundamental - ISTQB-CTFL - C/C++ by lotus IT Hub Pune. - BSNL BTS training by RTTC Pune.
2014-08 To 2016-09	Embedded SW Developer Micromax Instrument Pvt. Ltd. Pune, India Projects and Responsibilities Industrialized LED Display Board: - Firmware development to receive data over UART and transmit with different slave IDs for 8051 MCU. - Windows SW development using VB to test HW. - SMPS feedback circuitry design for 5V to 1.5 V output. - HW debugging and troubleshooting using logic Oscilloscope and multimeter. Stopwatch: - Written firmware for timer ISR (8051) to update data on 7 segment digits. - Developed application SW module to get data from keyboard and load into buffer. Automotive GPS clock: - PCB and hardware testing. - GPS L80 module output string verification over Hercules terminal. - Functional Testing.	Additional Abilities - Innovative and cost cutting point of view thinking. - Team Player
2011 To 2014	Education DKTE's Textile and Engineering Institute, An Autonomous Institute. Ichalkaranji, India - BE- Electronics and Telecom. Percentage- 67.94%	
2008 To 2011	Indira polytechnic, MSBTE Mumbai Board, India - Diploma- Electronics and Telecom. Percentage- 78.48%	
2008	Matriculation - Pune Board, India - SSC- Percentage- 70.76%	
	Projects 1. Campus wide communications using FM Technique. 2. Water Level Controller using IC 555. 3. Universal bank security system using microcontroller 8051. Hobby Projects: 1. RTOS-IOT based weather monitoring system (STM32F104RB). - Tools used; KeilV5, STM cube, STLink.	

